
a
MLANA protein fixed (118 aa)
15 choices
117 body codons
15 choices
5′UTR
AUG
CDS
UAA
3′UTR
A100
ROLE
METRIC
INPUT REGION
MFE (min)
Orthrus TE (max)
OPTIMIZE
CSI (max)
Ribosome load (max)
CONSTRAIN
Sequence feasibility
b
Orthrus TE ↑
CSI ↑
0.75
0.5
0.25
1
−MFE ↑
Ribosome load ↑
Initial
Best HV
5′UTR
AUG
CDS body
UAA
3′UTR
A100
Initial
0/117
BNT162b2
BNT162b2
MFE specialist
73/117
CYBA
HBB
TE specialist
69/117
CYBA
HBB
CSI specialist
75/117
CYBA
HBB
Ribosome-load specialist
71/117
CYBA
HBB
Best HV
72/117
CYBA
HBB
c
Pass feasibility filter
Feasible archive HV
Relative value
1.0
0.5
0.0
0
50
100
150
200
250
300
350
Generation
d
CAI
Orthrus TE
RiboNN
Orthrus half-life
CDS MFE
scanMiR
RiNALMo
Relative score
1.0
0.5
0.0
0
50
100
150
200
250
300
350
Generation